

Harpuneet Singh

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Science in Computer Science (Co-op)

Sep 2024 – May 2028

- **CGPA:** 3.90 | **Dean's List:** Fall 2024, Winter 2025, Fall 2025, Winter 2026
- **Coursework:** Data Structures & Algorithms, Database Systems, Object-Oriented Design, Software Engineering

TECHNICAL SKILLS

Languages: C/C++, Python, Java, SQL, JavaScript

Systems & Tools: Linux, Docker, Git, GitHub Actions, FastAPI, SQLite/WAL, Prometheus

Focus: Concurrency, Parsing, Fuzzing, Reliability, Networking Basics, CI/CD, Testing

EXPERIENCE

Student Developer

May 2026 – Present

Toronto Metropolitan University

Toronto, ON

- Built internal web systems for staff workflows; owned debugging, regression fixes, and iterative delivery in Agile cycles
- Hardened application behavior under real usage by reproducing failures, isolating root causes, and shipping targeted fixes
- Structured code for reuse so subsequent feature work required less duplication

Accounting Assistant

Jan 2026 – May 2026

HNM Logistics

Mississauga, ON

- Wrote Python automation over multi-thousand-row datasets to remove brittle manual reporting steps
- Added systematic validation to catch inconsistencies before operational close

FEATURED TECHNICAL PROJECTS

AEGIS — Fuzzing & Vulnerability Discovery Harness | *Python, C, Docker*

GitHub

- Engineered a mutation-based fuzzing harness against custom C/Python parsers with crash triage, deduplication, and minimized PoC inputs
- Identified and documented **3** unique lab defect classes (buffer overflow, null deref, logic/recursion) with reproducible artifacts
- Packaged fuzz/repro as a Dockerized pipeline with a thin adapter interface for new targets and CI smoke runs

FORGE — DAG Workflow Engine | *Python, FastAPI, asyncio, Docker,*

GitHub

Prometheus

- Designed a concurrent DAG workflow engine with dependency-aware scheduling, retries/backoff, and stuck-run recovery
- Sustained **7,000+** task completions/min under load tests with Prometheus instrumentation of queue depth and completions
- Implemented idempotent submission and dead-letter handling so exhausted failures isolate without stalling the scheduler

ATLAS — RAG Knowledge System + Eval Harness | *Python, FastAPI, BM25, TF-IDF*

GitHub

- Built a retrieval pipeline with hybrid BM25 + TF-IDF search and citation-backed answers; measured quality via a **54**-question eval harness
- Raised doc-precision from **0.52** to **0.62**; **47/54** eval cases passed